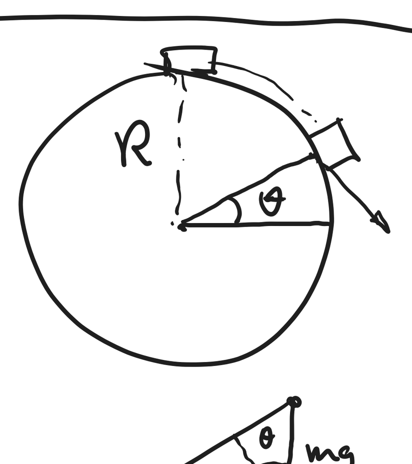
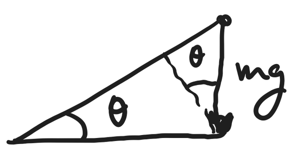
Problem 1) The work-energy theorem states that the initial release velocity would be : $v_o = \sqrt{2gl\cos\theta}$ where l represents the string length. For projectile motion, the horizontal distance traveled is

$$x = \frac{2v_o^2 \sin\theta\cos\theta}{g} = \frac{2(2gl\cos\theta)\sin\theta\cos\theta}{g}$$

$$\frac{dx}{d\theta} = 0 \Rightarrow \cos^3\theta - 2\cos\theta\sin^2\theta = 0 \Rightarrow \cos^2\theta = 2\sin^2\theta \Rightarrow \boxed{\tan\theta = \frac{1}{\sqrt{2}}}$$

Problem 2) The detachment occurs where the inward component of gravitational force becomes insufficient to represent the centripetal force. Let this point's angle be θ.





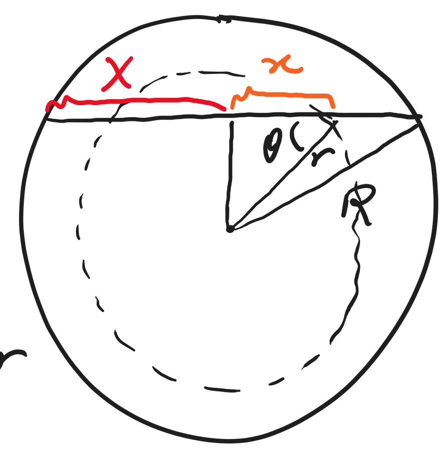
$$\Rightarrow F_c = \frac{mv^2}{R} = mg\sin\theta \quad \text{(I)}$$ two unknowns. But we also know from work-energy theorem that the vertical distance from the top to the detachment point is $R - R\sin\theta$.

$$\left.\begin{array}{l}\frac{1}{2}mv^2 = mgR(1-\sin\theta) \quad \text{(II)} \\ \text{(I)}\end{array}\right\} \Rightarrow 2mg(1-\sin\theta) = mg\sin\theta$$

$$\Rightarrow 3\sin\theta = 2 \Rightarrow \sin\theta = \frac{2}{3} \Rightarrow \boxed{\theta \simeq 0.7297 \text{ rad}}$$

Problem 3) 1) Where the ball reaches the top of earth center.

2) The gravitational force at radius r due to mass M_r inside a sphere of radius $r \le R$ is,



$$F_g = \frac{GM_r m}{r^2} = \frac{Gm(4\pi r^3\rho/3)}{r^2} = \frac{4}{3}\pi Gm\rho r$$

3) $F = mv\frac{dv}{dx} = -F_g\cos\theta = -\frac{4}{3}\pi Gm\rho x$

$$\Rightarrow \int_0^v mv\,dv = -\int_X^0 \frac{4}{3}\pi Gm\rho x\,dx \Rightarrow \frac{1}{2}mv^2 = \frac{4}{3}\pi Gm\rho X$$

$$\Rightarrow \boxed{v = \sqrt{\frac{8}{3}\pi G\rho X}}$$

Problem 4) The differential potential is $d\phi = -G\frac{dM}{r}$ where $dM = \rho\,dA = \rho\,2\pi x\,dx$.

$$\Rightarrow d\phi = -2\pi\rho G\frac{x\,dx}{r} = -2\pi\rho G\frac{x\,dx}{r} = -2\pi\rho G\frac{x\,dx}{(x^2+z^2)^{1/2}}$$

$$\Rightarrow \phi(z) = -\pi\rho G\int_0^a \frac{2x\,dx}{(x^2+z^2)^{1/2}} = -2\pi\rho G(x^2+z^2)^{1/2}\Big|_0^a$$

$$= -2\pi\rho G\left[(a^2+z^2)^{1/2} - z\right]$$

$F = -\nabla U = -m\nabla\phi$, problem is symmetric, therefore,

$$\boxed{F_z = -m\frac{\partial\phi}{\partial z} = 2\pi m\rho G\left[\frac{z}{(a^2+z^2)^{1/2}} - 1\right]}$$

Problem 5) The expected orbital speed v due to the galaxy mass M within radius R is,

$$\frac{GM}{r^2}m = \frac{mv^2}{R}$$

where m is the orbital mass. $\Rightarrow$ $v = \sqrt{\frac{GM}{R}}$

Because the luminous galactic mass ends beyond the galactic limbs, we would expect $v \propto \sqrt{\frac{1}{R}}$ when $R >$ limb length. However, we observe $v \propto$ const. which implies $\boxed{M(R) \propto R}$